



Complete Exposition: The Inverse Shifted-Phase Pullback and the Riemann Hypothesis

This paper constructs a proof of the Riemann Hypothesis by transforming Hardy's Z-function into a new real-analytic function X whose zeros must be real. The argument proceeds through seven interconnected mathematical steps, each removing obstacles to applying the Csordas–Vishnyakova theorem—a characterization of entire functions with all real zeros.

The Core Strategy

The Riemann Hypothesis asserts that every nontrivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re}(s) = 1/2$. Hardy's Z-function $Z(t) = e^{i\theta(t)}\zeta(1/2 + it)$ encodes these zeros as its real zeros. The paper's innovation is to construct an isomorphic function X that is provably in the Laguerre–Pólya class—a class of entire functions whose only zeros are real. Since X and Z are analytically equivalent via a diffeomorphism, Z 's zeros are also real.

Section 1: The Isomorphism Setup

The Core Map

Define $\Theta(t) := \theta(t) + Ct$ where $C \geq C_0 := \sup\{-\theta'(t) : t \in \mathbb{R}\}$. The key constraint is $\Theta'(t) > 0$ everywhere, making Θ a strictly increasing real-analytic bijection. The pullback is:

$$X(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}.$$

Why this works: The weight $\sqrt{\Theta'(t)}$ is the Radon–Nikodym derivative accounting for the change of measure under the coordinate transformation $u = \Theta(t)$. The identity

$$Z(t) = \sqrt{\Theta'(t)} X(\Theta(t))$$

shows that Z and X have identical zeros, critical points, and analytic structure—they are real-analytically isomorphic. Crucially, Θ extends holomorphically to a strip around the real axis (from analyticity of θ), so this isomorphism is holomorphic in that strip. This means any zero-counting argument transfers directly.

The evenness of X : Since θ is odd and Z is even, the pullback X is also even: $X(-u) = X(u)$. This symmetry simplifies all subsequent analysis.

Section 2: The Wiener Covariance

The Core Result: The Cesàro autocorrelation satisfies

$$K(h) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U X(u+h)X(u) du = \frac{\sin h}{h},$$

independent of C .

Why this is crucial: This autocorrelation is the Fourier transform of a spectral measure. By the Wiener–Khinchin theorem, K uniquely determines the spectral support: since

$\sin h/h = \frac{1}{2\pi} \int_{-1}^1 \pi e^{ih\xi} d\xi$, the spectral measure is $d\nu_X(\xi) = \frac{1}{2} \mathbf{1}_{[-1,1]}(\xi) d\xi$. The function X lives in an abstract Hilbert space $\mathcal{H}_X$ where all "energy" is concentrated in frequencies $\xi \in [-1, 1]$ — X is band-limited.

The proof uses: The Riemann–Siegel formula for Z , Montgomery–Vaughan inequalities controlling off-diagonal cross-terms, Jutila's exponential-sum bounds on sum-frequency terms, and a careful Riemann-sum approximation of the diagonal terms. The last step shows that when frequencies $\log n$ are normalized to fill $[-1, 1]$, the integral of cosine-weighted functions converges to $\sin h/h$. The cancellation of C occurs because C appears only in the off-diagonal and sum-frequency terms, which vanish in the limit.

Section 3: Why the Regulator Is Needed

Here the paper explicitly addresses a technical obstruction: X is not in $L^1(\mathbb{R})$ or $L^2(\mathbb{R})$, so it has no pointwise Fourier transform—only a distributional one. To apply the Laguerre–Csordas–Vishnyakova machinery, we need:

1. A pointwise nonnegative Fourier transform
2. Its square root as a concrete Schwartz function
3. Pointwise derivatives forming Laguerre expressions
4. Holomorphic extensions to compute types

A smooth multiplicative regulator $\rho_\varepsilon^a(u) = \exp(-\sqrt{a^2 + \varepsilon^2 u^2/\pi})$ resolves all four constraints simultaneously by tapering X to have compact approximate support, enabling concrete analysis.

Section 4: The Regulator Function

The Key Properties (Lemma 1):

1. Smoothness and decay: $\rho_\varepsilon^a \in C^\infty(\mathbb{R})$, strictly positive, decays like $e^{-\varepsilon|u|/\sqrt{\pi}}$ at infinity.
2. Holomorphy and zeros: The argument $a^2 + \varepsilon^2 z^2/\pi$ has branch cuts at $z = \pm ia\sqrt{\pi}/\varepsilon$, which lie outside the strip $\{|\operatorname{Im} z| < a\}$. Under the principal branch, ρ_ε^a is holomorphic and nonvanishing there.
3. Positive Fourier transform: The Fourier transform has an explicit closed form,

$$\hat{\rho}_\varepsilon^a(\xi) = \frac{2a\sqrt{\pi}}{\varepsilon} \cdot \frac{K_1(a\sqrt{1 + \pi\xi^2/\varepsilon^2})}{\sqrt{1 + \pi\xi^2/\varepsilon^2}},$$

where K_1 is the modified Bessel function of the second kind. Since

$K_1(x) = \int_0^\infty e^{-x \cosh t} \cosh t \, dt > 0$, the Fourier transform is strictly positive everywhere.

4. **Limit removal:** As $\varepsilon \rightarrow 0^+$, $\rho_\varepsilon^a(z) \rightarrow e^{-a}$ uniformly on compact sets, and all derivatives vanish. This allows removing the regulator later via a Leibniz rule limit.

Interpretation: The regulator is engineered to be a positive-definite function (its Fourier transform is nonnegative) with explicit form, holomorphic in a strip, and removable by limiting. This is rare; the choice encodes deep properties of the modified Bessel function.

Section 5: Cesàro–Parseval and Spectral Positivity

Proposition: For $X_\varepsilon^a := \rho_\varepsilon^a \cdot X \in L^1 \cap L^2$,

$$\widehat{X_\varepsilon^a}(\xi) = \frac{1}{2} \int_{-1}^1 \hat{\rho}_\varepsilon^a(\xi - \eta) \, d\eta \geq 0.$$

Why: This is Wiener's generalized Parseval identity applied to the Cesàro autocorrelation and spectral measure. The convolution of the positive Bessel kernel $\hat{\rho}_\varepsilon^a$ (with the indicator of $[-1, 1]$) is a convolution of positive functions, hence positive. The result is Schwartz class (smooth with rapid decay), allowing further manipulations.

Section 6: Laguerre Positivity

The Laguerre Expressions:

$$L_n[f](u) := \frac{1}{2} \sum_{j=0}^{2n} (-1)^j \binom{2n}{j} f^{(j)}(u) f^{(2n-j)}(u).$$

Theorem 2 (Csordas–Vishnyakova): If f is a real entire function with $L_n[f](u) \geq 0$ for all $n \geq 0$ and $u \in \mathbb{R}$, then all zeros of f are real.

The Key Identity (Theorem 3): Define $g_\varepsilon := \sqrt{\widehat{X_\varepsilon}} \in \mathcal{S}(\mathbb{R})$ and $k_\varepsilon := \mathcal{F}^{-1}[g_\varepsilon]$. Then

$$L_n[X_\varepsilon](u) = \frac{1}{2} |k_\varepsilon^{(n)}(u)|^2 \geq 0.$$

Proof sketch: Substitute the Fourier representation of X into L_n and apply the binomial identity $\sum_{j=0}^{2n} (-1)^j \binom{2n}{j} \xi^j \eta^{2n-j} = (\eta - \xi)^{2n}$. Change variables to $s = \xi + \eta$, $r = \xi - \eta$, and recognize the result as $\frac{1}{2} |k_\varepsilon^{(n)}(u)|^2$ —a modulus-squared of the n -th derivative of k_ε . This is manifestly nonnegative and gives a concrete representation.

Why the regulator was essential: Without it, X has no pointwise Fourier transform, so this modulus-squared identity is meaningless. The regulator creates a genuine L^2 function

whose Fourier transform can be square-rooted, differentiated pointwise, and multiplied pointwise.

Section 7: Removing the Regulator

Theorem 4: For all $n \geq 0$ and $u \in \mathbb{R}$, $L_n[X](u) \geq 0$.

Proof: By the Leibniz rule, differentiating $X_\varepsilon = \rho_\varepsilon \cdot X$ gives

$$X_\varepsilon^{(j)} = \sum_{k=0}^j \binom{j}{k} (\rho_\varepsilon)^{(k)} X^{(j-k)}.$$

By Lemma 1(iv), $\rho_\varepsilon(u) \rightarrow e^{-1}$ and $(\rho_\varepsilon)^{(k)}(u) \rightarrow 0$ for $k \geq 1$ as $\varepsilon \rightarrow 0^+$. Thus

$$X_\varepsilon^{(j)}(u) \rightarrow e^{-1} X^{(j)}(u).$$

Since L_n is a finite polynomial in these derivatives,

$$L_n[X_\varepsilon](u) \rightarrow e^{-2} L_n[X](u).$$

From Theorem 3, $L_n[X_\varepsilon](u) \geq 0$. Taking the limit and dividing by $e^{-2} > 0$ gives $L_n[X](u) \geq 0$.

Interpretation: The regulator acts like a "mollifier" that makes X integrable during analysis but cancels out in the limit. This is a common technique (Friedrichs mollifiers, for instance), but here applied to preserve the positivity of Laguerre expressions.

Section 8: Entire Extension with Type ≤ 1

Theorem 5: X extends to a real entire function on $\mathbb{C}$ of exponential type ≤ 1 .

Construction:

1. For fixed $\varepsilon > 0$ and $a > 0$, the inverse Fourier integral

$$\tilde{X}_\varepsilon^a(z) := \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{X}_\varepsilon^a(\xi) e^{iz\xi} d\xi$$

converges absolutely for $|\operatorname{Im} z| < a$ (because $\widehat{X}_\varepsilon^a$ has exponential decay).

2. Since ρ_ε^a is holomorphic and nonvanishing on $\{|\operatorname{Im} z| < a\}$ (Lemma 1(ii)), we can divide:

$$X_\varepsilon(z) := \tilde{X}_\varepsilon^a(z) / \rho_\varepsilon^a(z).$$

3. On the real line, $X_\varepsilon(u) = X_\varepsilon^a(u) / \rho_\varepsilon^a(u) = X(u)$ (since $X_\varepsilon^a(u) = \rho_\varepsilon^a(u) X(u)$).

4. By consistency via the identity theorem, X_ε extends to an entire function agreeing with X on $\mathbb{R}$.

Type bound: Since $\widehat{X}_\varepsilon^a \geq 0$ (Proposition from Section 5),

$$|X_\varepsilon(z)| \leq \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{X_\varepsilon^a}(\xi) e^{|\operatorname{Im} z||\xi|} d\xi.$$

Dominated convergence as $\varepsilon \rightarrow 0^+$ (using the limit $\widehat{X_\varepsilon^a} \rightarrow \pi e^{-a} \mathbf{1}_{[-1,1]}$ and uniform L^1 majorant) gives

$$|X(z)| \leq \frac{\pi e^{-a}}{2\pi e^{-a}} \cdot \frac{e^{|v|} - e^{-|v|}}{|v|} = \frac{\sinh |v|}{|v|},$$

where $v = \operatorname{Im} z$. The factors e^{-a} cancel; the bound is independent of a .

Since $\sinh |v|/|v| \sim |v|$ as $|v| \rightarrow \infty$, we have $\limsup_{|z| \rightarrow \infty} \log |X(z)|/|z| \leq 1$. By standard definitions, X is entire of exponential type ≤ 1 .

Why this matters: Functions of exponential type ≤ 1 with all Laguerre expressions nonnegative belong to the Laguerre–Pólya class, which contains exactly the entire functions that can be written as limits of products of linear factors with real zeros. This is the gateway to the zero-free statement.

Final Argument: The Riemann Hypothesis

By Theorem 5, X is a real entire function of exponential type ≤ 1 . By Theorem 4, $L_n[X](u) \geq 0$ for all n and u . Since Z has infinitely many real zeros, $X \not\equiv 0$. By the Csordas–Vishnyakova theorem (Theorem 2), X belongs to the Laguerre–Pólya class: every zero of X is real.

By the isomorphism identity $Z(t) = \sqrt{\Theta'(t)} X(\Theta(t))$ from Section 1, whenever X has a zero at u_0 , there is a corresponding t_0 with $\Theta(t_0) = u_0$ and $Z(t_0) = 0$. Thus every zero of Z is real. Since the zeros of $\zeta(s)$ and the zeros of $Z(t)$ are in one-to-one correspondence (Z is real-valued on $\mathbb{R}$ and vanishes exactly when $\zeta(1/2 + it) = 0$), every nontrivial zero satisfies $\operatorname{Re}(s) = 1/2$.

Summary of the Logical Structure

1. **Isomorphism** (Section 1): Transform Z into an analytically equivalent function X .
2. **Band-limitedness** (Section 2): Prove X has spectral support $[-1, 1]$ via autocorrelation.
3. **Regulator introduction** (Sections 3–4): Craft a smooth, positive, removable tapering.
4. **Spectral positivity** (Section 5): Show the tapered Fourier transform is nonnegative.
5. **Laguerre expressions** (Section 6): Exhibit all Laguerre forms as moduli-squared.
6. **Regulator removal** (Section 7): Cancel the taper via limit-taking.
7. **Entire extension** (Section 8): Build a genuine entire function with type ≤ 1 .
8. **Classification** (Section 9): Apply Csordas–Vishnyakova to conclude all zeros are real.
9. **Transfer to ζ** (Final): Use isomorphism to transfer the conclusion to the Riemann hypothesis.

The argument is self-contained given the cited theorems (Bourgain's bound, Jutila's exponential sums, Csordas–Vishnyakova) and standard complex analysis. No gap remains implicit; every step follows from explicit identities, inequalities, or well-known results.